Kou Model

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1 Introduction

This project discusses and implements a pricing and hedging framework based on the "Kou Model", introduced by Steven Kou in [1], a model which allows for sudden jumps. Delta-hedging is a popular hedging strategy used to reduce the directional dependence of a portfolio on the value of the underlying stock. When applied to the Black-Scholes model, in principle, Delta-hedging serves its purpose reasonably well. The aim of this project is to see whether it also does so when applied to the Kou-Model.

2 Kou-Model

The Kou-model is an extension of the Black-Scholes model that allows for sudden jumps/falls in the (stochastic) time-evolution of the log-returns of a given stock, in addition to the brownian evolution predicted by the Black-Scholes model. The main difference between the Kou-model and the similar Merton-model, is that the Kou-Model assumes that the random jumps are drawn from

a double, asymmetric exponential distribution, as opposed to a normal distribution. The reasoning behind the asymmetry this is that in general humans tend to overestimate negative events, and underestimate positive events, hence more extreme negative jumps are likelier to occur than extreme positive jumps. Further more, the choice of an exponential distribution as opposed to a normal distribution means that more extreme jumps are likelier to occur in the Kou-model than the Merton model, which better matches real stock-data. When compared to practical data, the Kou-model is known to produce a better fit to volatility smiles as well.

2.1 Time evolution according to the Kou Model

Let $S(t)$ be the value of a certain stock at time t, and S_0 be the value at a given initial time. Defining the log return $X_t \equiv \log(S(t)/S_0)$, the Kou model assumes the following time evolution of $S(t)$.

$$X(t) = (r - \frac{\sigma^2}{2} - \lambda\kappa)t + \sigma W_t + \sum_{i=1}^{N(t)} Y_i \tag{1}$$

The meaning of each of the symbols on the right hand side of Eq. (1) is given in the following table :

Symbol	Meaning
r	risk-free interest rate
σ	volatility of stock
W_t	standard brownian motion
$N_t \sim$ Poisson (λt)	number of jumps up to time t
Y_i	iid jump sizes with double exponential distribution
$\kappa = E\left[e^Y\right] - 1$	mean jump multiplier

More specifically, W_t,$N(t)$ and Y_i's are all random variables with the following distributions.

- W_t, standing for standard Brownian evolution, is drawn from a normal distribution.
- $N(t)$ stands for number of jumps that can occur upto time t, and is drawn from a Poisson distribution with mean number of jumps given by (λt).
- Y_i are the variables corresponding to the jump sizes themselves, and are drawn from an asymmetric double exponential distribution that is characterized by the two decay rates η_1, η_2. Essentially, upward jumps are drawn from an Expontial (η_1) distribution and downward jumps from an Exponential (η_2).

2.2 Option Pricing : The Fourier Transform Method

A european call option with strike price K, initial value S_0 and an expiry time T, has a fair market price of :

$$C = e^{-rT} E[\mathrm{Max}\big(S(T) - K, 0\big)] \tag{2}$$

The option prices can be numerically in a highly efficient manner using the FFT method introduced by Carr and Madan [2], who showed that one can obtain the option prices for a model using a highly efficient numerical fourier transform, as long as the characteristic function was known. The underlying theory is as follows :

Defining the log-moneyness $x \equiv \log(K/S_0)$, let us view the scaled option price $c = C/S_0$ as a function of x for a fixed value of r, T.

$$c_{r,T}(x) = \frac{C_{r,T}}{K_0}. \tag{3}$$

Usually, it is not possible to perform a fourier transform on the function $c_{r,T}(x)$, as it does not vanish as $x \to -\infty$, instead it tends to 1. However, we can instead consider the damped call price $e^{\alpha x} c_{r,T}$, where $\alpha > 0$. This damped function is in-fact fourier transformable, and it can be shown that it's fourier transform is the so-called "Carr-Madan" function $M_{r,T}$ given by :

$$M_{r,T}(v) = \frac{\mathrm{e}^{-rT} \phi_T(v - (\alpha + 1)\mathrm{i})}{\alpha^2 + \alpha - v^2 + \mathrm{i}(2\alpha + 1)v}. \tag{4}$$

Here, $\phi_T(u)$ is the characteristic function of the Kou-process, defined as :

$$\phi_T(u) = E[e^{iu \log\left(X(T)\right)}]. \tag{5}$$

which can be computed analytically, for which a derivation is laid out in section 2.2.1. Once we obtain an expression for the Carr-Madan function $M_{r,T}$, it can be inverse-Fourier transformed numerically to yield the call price, yielding :

$$c_{r,T}(x) = \frac{\exp(-\alpha x)}{2\pi} \int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{i}vx} M_{r,T}(v) \mathrm{d}v. \tag{6}$$

Note that we multiply the inverse-Fourier transform by a factor of $e^{-\alpha x}$ to compensate for the damping factor we had initially introduced.

2.2.1 Characteristic function ϕ_T of the Kou process

The characteristic function $\phi_T(u)$ for the kou process is defined as :

$$\phi_T = E[e^{iuX_T}] \tag{7}$$

We can find ϕ_T for the kou process as follows. The kou process is a levy process, and hence has a levy exponent of the process, $\Psi(z)$ defined as

$$\psi(z) = \frac{1}{T} \log E[e^{zX_T}], \tag{8}$$

If we have an analytic form for $\psi(z)$ then one can compute $\phi_T(u)$ exactly in terms of ψ :

$$\phi_T(u) = e^{T\psi(iu)}. \tag{9}$$

For finding the analytic form for $\psi(z)$ for the kou process, we break $\frac{1}{T}E[e^{zX_T}]$ down as follows :

$$E[e^{zX_T}] = \log\left((E[e^{(r-\frac{\sigma^2}{2}-\lambda\kappa)zT}]\right) + \log\left(E[e^{\sigma W_T}]\right) + \log\left(E[e^{\sum_{i=1}^{N(T)} Y_i z}]\right) \tag{10}$$

The first term contains deterministic variables, and is hence simply

$$\log\left((E[e^{(r-\frac{\sigma^2}{2}-\lambda\kappa)zt}]\right) = (r - \frac{1}{2}\sigma^2 - \lambda\kappa)zT. \tag{11}$$

The second term is brownian motion, hence it is simply the log-MGF of the normal distribution given by :

$$\log\left(E[e^{\sigma W_T}]\right) = \frac{1}{2}\sigma^2 z^2 T \tag{12}$$

For the third term, since $N(t)$ is a poisson variable with mean frequency λT, it has a log MGF of the following form :

$$\log\left(E[e^{\sum_{i=1}^{N(T)} zY_i}]\right) = \lambda T \times (E[Y \cdot z] - 1) \tag{13}$$

and $E[Y]$ is known as Y is drawn from an assymetric exponential distribution, where the exponential distributions for upward and downward jumps are parametrized by η_1, η_2 respectively, with the probability of upward jumps being p and downward jumps being $1-p$. Hence :

$$E[Y] = \frac{p\eta_1}{\eta_1 - z} + \frac{(1-p)\eta_2}{\eta_2 + z} \tag{14}$$

$$\psi(z) = (r - \frac{1}{2}\sigma^2 - \lambda\kappa)z + \frac{1}{2}\sigma^2 z^2 + \lambda\left(\frac{p\eta_1}{\eta_1 - z} + \frac{(1-p)\eta_2}{\eta_2 + z} - 1\right) \tag{15}$$

3 Calculation of Δ

Δ is defined as $\frac{\partial C}{\partial S_0}$. One of the nice things about the fourier-transform method is that is a simple trick allows us to evaluate $\Delta_{r,T}$ as a function of $x = \log(K/S_0)$ without explicitly numerically differentiating the option prices we found. The idea is as follows. First we write down the call price C in terms of the Carr-Madan Function.

$$C = \frac{S_0}{2\pi} \int_{-\infty}^{\infty} e^{-i(\alpha+iv)x} M(v) dv \tag{16}$$

Taking the derivative with respect to S_0, and noting that the dependence on S_0 comes from an explicit factor of S_0 in the numerator and from the $x = \log(K/S_0)$ within the integral, we have :

$$\frac{\partial C}{\partial S_0} = \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i(\alpha+iv)x} M(v) dv \right) + S_0 \cdot \frac{1}{\pi} \int_{0}^{\infty} \frac{\partial}{\partial S_0} \left[e^{-(\alpha+iv)x} \right] M(v) dv. \tag{17}$$

From the definition of x, we can show that

$$\frac{\partial e^{-(\alpha+iv)x}}{\partial S_0} = e^{-(\alpha+iv)x} \cdot \frac{\alpha + iv}{S_0}. \tag{18}$$

Thus piecing it all together gives us :

$$\Delta = \frac{e^{-\alpha x}}{2\pi} \int_{-\infty}^{\infty} e^{-ivx} (1 + \alpha + iv) M(v) dv. \tag{19}$$

Hence, in order to compute Δ, (something we need in order to simulate delta-hedging for the kou model), we do not need to numerically differentiate the option prices, we can simply do a numerical inverse fourier transform of the Carr-Madan function multiplied by a factor of $(1 + \alpha + iv)$. In practice, to calculate Δ while performing Monte Carlo Simulations, we first construct a surface(grid) of $\Delta_r(x, T)$ as a function of the log-moneyness x and expiry time T, (for a fixed rate r), and interpolate within this grid to quickly evaluate Δ at any K, S_0, T.

4 Validating Option prices obtained via the FFT with Monte Carlo results

While the FFT does provide an extremely efficient way to generate option prices and deltas quickly for range of strike prices and times, we benchmark the results against the option prices obtained by performing monte-carlo simulations, to verify that it does indeed provide accurate option prices. As we can see in Fig. (2), the FFT method produces accurate option prices when compared against monte-carlo, and unlike the monte-carlo, it converges much faster.

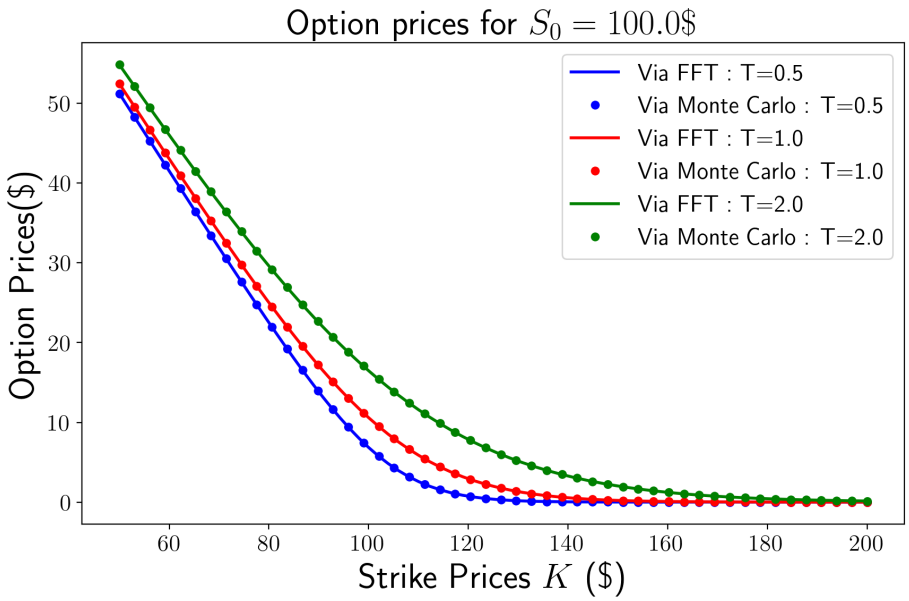

Figure 1: Here we compare the option prices obtained via the FFT to the Option prices obtained via the Monte Carlo simulations. The spot price was fixed at $S_0 = 100\$$ and calculations were performed for $T = 0.5, 1.0, 2.0$ respectively. The parameters used for these simulations were $r = 0.05, \sigma = 0.15, \lambda = 1.5, \eta_1 = 25.0, \eta_2 = 10.0$.

5 Delta-Hedging

For the black-scholes process, delta-hedging is a well known hedging technique one applies to a portfolio in order to make the value of the portfolio directionally neutral of the underlying stock price and reduce the "fat-tails" in portolio values one obtains as the market moves.

The idea is that for every call option that the portfolio contains, it also assumed a "hedged" position simultaneously that shorts exactly Δ of the same stock (here Δ is the rate of change of the call option price w.r.t the underlying stock price). This ensures that the overall portfolio remains independent of the underlying stock price. Mathematically, one would attempt to adjust the number continously as the stock evolves, in practice, Δ is calculated periodically at a fixed (small) time interval and the number of shorted stock is adjusted accordingly.

While this process works well for the Black-Scholes process, it is less effective when applied to the Kou-process. This stems from the fact that the kou process incorporates "sudden jumps" in the stock price, and while continous hedging

works well against infinitesimal, or small changes in the stock price, it doesn't work well against large jumps, as one cannot adjust hedges rapidly enough to match new prices, and this leads to "slippage".

This is illustrated in the following set of figures, where we compare values of a delta-hedged portfolio, while looking at three different values of λ, the average number of "jumps" in a unit time interval. We note that $\lambda = 0$ corresponds

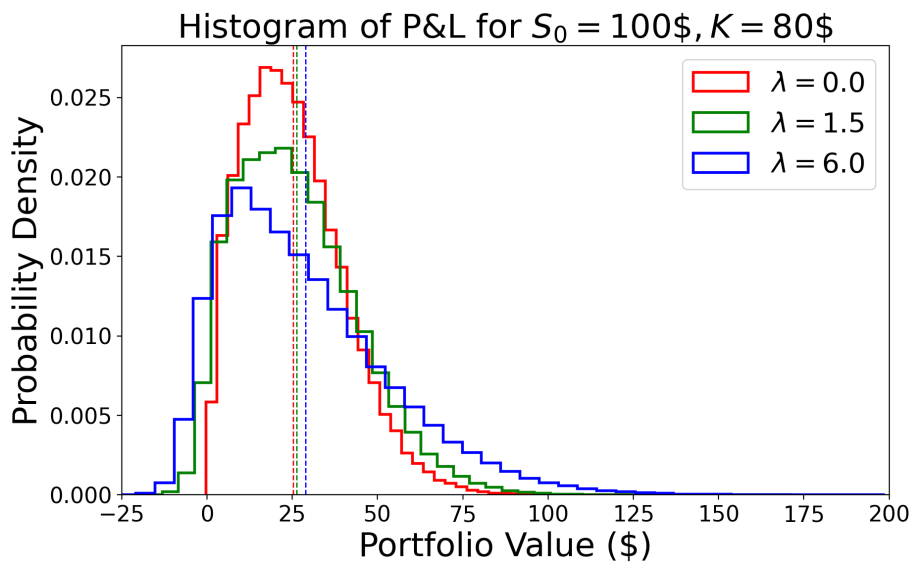

Figure 2: Here we compare the values of delta-hedged portfolios. The spot price was fixed at $S_0 = 100\$$, strike price at $K = 80\$$ and time to expiry was $T = 1$ (year). The parameters used for these simulations were $r = 0.05, \sigma = 0.15, \eta_1 = 25.0, \eta_2 = 10.0$, while λ was selected as $0.0, 1.5, 6.0$ respectively. The hedged position was adjusted twice every working day. The vertical dashed line represents the mean

to a perfect black-scholes process, hence delta-hedging works well, and helps produce a distribution more narrowly centered around the mean. For $\lambda = 1.5$, we see this is less true, as the distribution is wider, even extending well within the negatives. This is even more pronounced for $\lambda = 6.0$, where the distribution is still wider.

The conclusion that can be drawn from this is that in practice,delta-hedging is an imperfect process that doesn't necessarily work as desired when applied to more realistic models than the black-scholes model, such as the Kou Model considered in this project.

References

[1] S. G. Kou, "A jump-diffusion model for option pricing," *Management science*, vol. 48, no. 8, pp. 1086–1101, 2002.

[2] P. Carr and D. Madan, "Option valuation using the fast fourier transform," *Journal of computational finance*, vol. 2, no. 4, pp. 61–73, 1999.